

AI Engineer Assessment

Round 1 — Take-Home Project

SUBMISSION DEADLINE



3 DAYS (72 HOURS)

from the time this assignment is shared.

1 PROBLEM STATEMENT

Analytics Vidhya is building AI-native learning tools for the data science community. For this assessment, you will build a **Python Programming Q&A Assistant** — an AI-powered system that helps data science learners get accurate, grounded answers to their Python questions.

Build a question-answering system using the provided dataset that is able to answer any Python-related query asked by AI learners. You should host this app somewhere publicly accessible. It is preferred that the app uses **FastAPI** as the backend framework. You should also perform proper testing of your API.

2 DATASET TO BE USED

Stack Overflow — Python Questions & Answers

<https://www.kaggle.com/datasets/stackoverflow/pythonquestions>

3 WHAT YOU NEED TO BUILD

01 RAG Pipeline or Agent

Build a retrieval-augmented generation pipeline or agentic system using LangChain, LlamaIndex, LangGraph, or any framework of your choice.

02 REST API with FastAPI

Wrap your pipeline in a FastAPI service. At minimum, expose a `POST /ask` endpoint that accepts a question and returns a grounded answer, and a `GET /health` endpoint.

03 API Testing

Test your API against a set of diverse Python-related queries. Document the questions, responses, and any failure cases or edge cases you observed.

04 **Deployment** BONUS

Deploy the application to a publicly accessible platform such as Hugging Face Spaces, Render, Railway, or any free-tier cloud. Include the live URL in your README.

4 WHAT YOU NEED TO SUBMIT

- ✓ **GitHub Repository** — Public repo with clean code, a README with setup instructions, and a `.env.example` for required environment variables.
- ✓ **Test Results** — A file or notebook showing at least 8 test queries with their responses and observations on quality.
- ✓ **Slide Deck** (max 10 slides) — What you built, key design decisions, architecture diagram, and how you would scale this for 100+ concurrent users (latency, async calls, database, caching, cost).
- ✓ **Deployed App URL** BONUS — Live link to the hosted application included in your README.

5 NON-DISCLOSURE

This assessment is **strictly confidential** and is shared with you solely for the purpose of evaluating your candidacy for the AI Engineer role at Analytics Vidhya.

Do not share this assignment, its contents, or your solution with anyone else — including publicly on GitHub, forums, social media, or any other platform — until the hiring process has formally concluded.

Sharing this document or submitting a solution that has been developed with undisclosed external assistance will result in immediate disqualification from the hiring process.